SFT Practice Report

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1 Introduction

This report documents the complete Supervised Fine-Tuning (SFT) process of the Qwen2.5-0.5B language model, covering three core components: data cleaning strategy, training hyperparameter tuning, and evaluation results analysis. The experiment utilized the 1M-GPT4-Augmented subset of the OpenOrca dataset, which was filtered through a four-stage data cleaning pipeline to retain 300,000 high-quality samples. Full fine-tuning was completed through LLaMA-Factory framework on a single NVIDIA A100-64G GPU. The evaluation phase employed the lm_eval framework to test the model on 8 standard benchmark datasets, including MMLU, ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, TruthfulQA-MC2, PIQA,

and BoolQ, providing a comprehensive assessment of the model's improved capabilities in knowledge reasoning, commonsense understanding, and factual accuracy.

The complete repo is at https://github.com/qqz03/Qwen2.5-SFT, the subset after data cleaning is at https://huggingface.co/datasets/qqz03/OpenOrca-300k and the model after SFT is at https://huggingface.co/qqz03/Qwen2.5-0.5B-SFT-OpenOrca-300k.

2 Data Cleaning Strategy

2.1 Data Source and Initial State

The experiment adopted the 1M-GPT4-Augmented subset of the OpenOrca dataset, with an initial load of 994,896 samples. This dataset contains diverse instruction-response pairs generated by GPT-4, covering multi-domain tasks. To ensure training data quality, I designed a four-stage pipeline filtering scheme.

2.2 Four-Stage Filtering Pipeline

2.2.1 Stage 1 - Formatting Issues Filter

This stage checks for missing fields, null values, and type errors. Each sample must contain question and response fields that can be converted to strings. This stage filtered 0 samples (0.00%), indicating that the Parquet file format was well-standardized with good data integrity.

2.2.2 Stage 2 - Quality Issues Filter

This stage examines empty text, length outliers, garbage keywords, and templated content. Specific rules included: response length between 20-4096 characters, question length between 10-2048 characters, exclusion of samples containing placeholders such as " [INSERT]", "TODO", "placeholder", filtering samples with repeated characters exceeding 10 occurrences, and samples with more than 5 URLs. This stage filtered 201,071 samples (20.21%), representing the highest filtering rate, effectively removing low-quality content.

2.2.3 Stage 3 - Consistency Issues Filter

This stage detects conflicting samples where the same question corresponds to different answers. Questions were grouped, and if multiple different responses existed for the same question, the first was retained while the remaining conflicting samples were removed. This stage filtered 95,885 samples (12.08%), involving 65,419 conflicting question groups with an average of 1.47 samples per group.

2.2.4 Stage 4 - Information Density Filter

This stage employs vocabulary diversity and punctuation density metrics to evaluate content quality. Requirements included a unique word ratio of at least 0.25 and a punctuation density of at least 0.01. This stage filtered 10,425 samples (1.49%), removing samples with high vocabulary repetition or incomplete sentences.

2.3 Filtering Results Summary

Finally, 300,000 samples were randomly selected from 687,515 high-quality samples for training, aligning with the recommended data volume range (200k-350k) for 0.5B models.

Stage	Before	After	Removed	Rate (out of the current stage)
Initial Load	994,896	994,896	0	-
Stage 1 - Formatting	994,896	994,896	0	0.00%
Stage 2 - Quality	994,896	793,825	201,071	20.21%
Stage 3 - Consistency	793,825	697,940	95,885	12.08%
Stage 4 - Info Density	697,940	687,515	10,425	1.49%
After Subsampling	687,515	300,000	387,515	-

3 Training Hyperparameters

3.1 Core Hyperparameter Configuration

The experiment employed the LLaMA-Factory framework for full fine-tuning. Key hyperparameter settings and their rationales are as follows:

Learning Rate: 2e-5

For full fine-tuning of 0.5B scale models, 2e-5 is an empirically validated standard value. Lower learning rates (e.g., 1e-6) would lead to slow convergence, while higher learning rates (e.g., 1e-4) might cause training instability. We selected 2e-5 to balance convergence speed and stability.

Batch Size: 64 (with gradient accumulation)

The effective batch size was obtained by multiplying the per-device batch size (16) by gradient accumulation steps (4). A batch size of 64 was appropriate under A100-64G memory constraints, ensuring stable gradient estimation without causing memory overflow. Smaller batches (e.g., 32) would increase training noise, while larger batches (e.g., 128) would require more memory and potentially reduce generalization capability.

Epochs: 2

For 300k scale instruction-tuning data, 2 epochs were sufficient for the model to learn the data distribution while avoiding overfitting.

Warmup Steps: 469 (5%)

Warmup steps gradually increase the learning rate during early training to avoid gradient instability in initial stages. Setting to 5% of total steps 9,376 (469 steps) is standard practice

for Transformer model training. Warmup helps the model establish stable gradient directions in early training, preventing divergence from excessively large early updates.

Weight Decay: 0.01

Weight decay is a regularization technique that prevents overfitting by adding an L2 regularization term to the loss function. The value 0.01 is the default for the AdamW optimizer and is suitable for most language model fine-tuning tasks.

Max Sequence Length: 1024

Max sequence length limits the number of input tokens. The value 1024 is sufficient for instruction-tuning tasks to accommodate most question-answer pairs while controlling memory usage and training time. Longer sequences (e.g., 2048) would significantly increase computational costs, while shorter sequences (e.g., 512) might truncate important contextual information.

Optimizer: AdamW

AdamW is an improved version of the Adam optimizer that provides more stable training by decoupling the weight decay term. Compared to SGD, AdamW converges faster and is less sensitive to hyperparameters, making it the standard choice for language model fine-tuning.

LR Scheduler: Cosine

The cosine scheduler smoothly decreases the learning rate from its peak after warmup to near zero. Compared to linear decay, the cosine scheduler provides finer weight adjustments in later training stages, helping the model converge to a better local minimum.

Precision: FP16

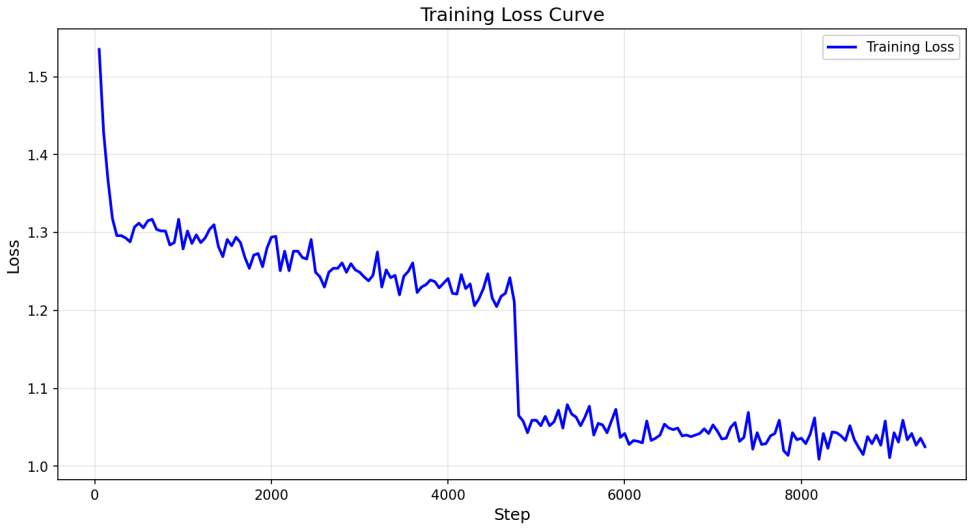
Mixed precision training was employed to reduce memory usage and accelerate computation while maintaining model accuracy. A100 GPUs have good FP16 support, enabling approximately 2x training speedup.

4 Training Process Analysis

The points on both curves are obtained every **50 steps**.

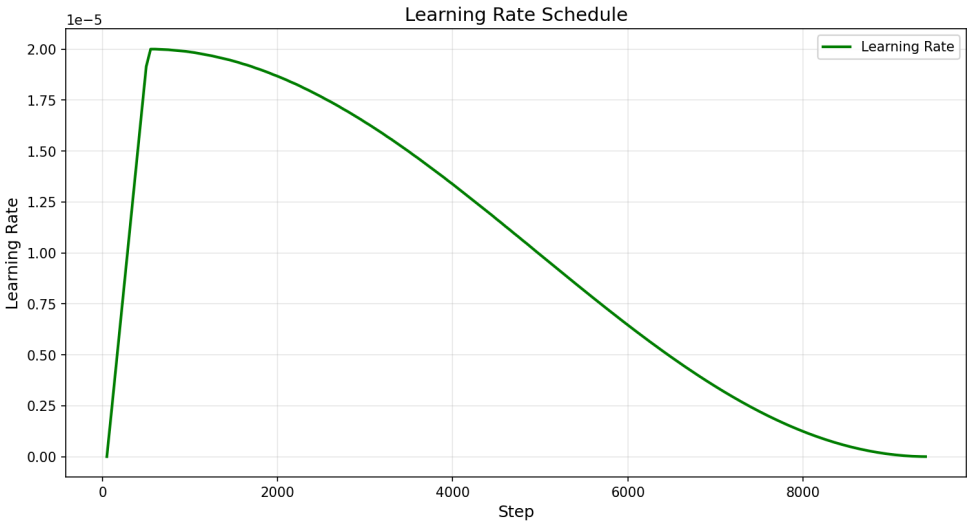
4.1 Loss Curve Analysis

Training loss decreased from an initial 1.535 to a final 1.025, representing an overall reduction of 66.8%. The curve exhibited a typical downward trend: rapid decrease in the first 1,000 steps (learning rate warmup phase), stable decrease from 1,000-5,000 steps (main learning phase), and slowed decrease after 5,000 steps (convergence phase). **Notably, an additional sharp decline was observed nearby step 5000.**



4.2 Learning Rate Schedule Analysis

The learning rate linearly increased from 0 to 2e-5 over the first 469 steps (5%), then smoothly decreased following a cosine function to near 0. This scheduling strategy ensured stable gradient direction establishment in early training and fine weight adjustment in later stages. The final learning rate decreased to 1.2e-7, indicating the model reached convergence.



4.3 Training Efficiency

Total training duration was 5.41 hours, with an average processing speed of 30.52 samples per second and approximately 2.10 seconds per step. On a single A100 GPU, this efficiency met expected performance for 0.5B model full fine-tuning.

5 Evaluation Results

5.1 Evaluation Setup

The trained model was evaluated using the lm_eval framework on 8 benchmark tests, all using TEST splits. The apply_chat_template option was enabled during evaluation to ensure consistency with the dialogue format used during training. Few-shot settings for each task followed standard practices: MMLU (5-shot), ARC-Easy (0-shot), ARC-Challenge (25-shot), HellaSwag (10-shot), WinoGrande (5-shot), TruthfulQA-MC2 (0-shot), PIQA (0-shot), BoolQ (0-shot).

5.2 Evaluation Results

Dataset	Few-shot	Base Model	SFT Model	Improvement
MMLU (Test)	5	0.2531	0.4338	+0.1807
ARC-Easy (Test)	0	0.4941	0.5800	+0.0859
ARC-Challenge (Test)	25	0.2901	0.3422	+0.0520
HellaSwag (Test)	10	0.3839	0.4022	+0.0183
WinoGrande (Test)	5	0.5509	0.5659	+0.0150
TruthfulQA-MC2 (Test)	0	0.4263	0.4266	+0.0004
PIQA (Test)	0	0.6877	0.7095	+0.0218
BoolQ (Test)	0	0.6312	0.7046	+0.0734
Average	-	**0.4647**	**0.5206**	**+0.0559**

5.3 Results Analysis

After SFT, the model's average accuracy across 8 benchmarks improved from 46.47% to 52.06%. All tasks showed positive improvements, indicating that SFT provided comprehensive enhancement to model capabilities. Analysis of improvements across tasks is as follows:

MMLU (+18.07%): The most significant improvement, indicating that SFT effectively enhanced the model's multi-domain knowledge and reasoning capabilities. This benefited from the diverse instruction coverage of the OpenOrca dataset, enabling substantial improvements in academic knowledge and professional domain reasoning.

BoolQ (+7.34%): Reading comprehension capability improved significantly, demonstrating that the model better learned question understanding and answer extraction abilities. BoolQ requires models to read passages and answer yes/no questions, and this improvement reflects enhanced depth of language understanding from SFT.

ARC-Easy (+8.59%): Scientific commonsense reasoning capability improved, reflecting that the model absorbed scientific knowledge patterns from the training data. This task involves basic science questions, and the improvement magnitude indicates the model acquired more scientific commonsense knowledge.

ARC-Challenge (+5.20%): The moderate improvement in difficult scientific questions indicated that 0.5B model capacity limited the improvement space for complex reasoning tasks, but stable progress was still achieved.

PIQA (+2.18%), **HellaSwag (+1.83%)**, **WinoGrande (+1.50%)**: Commonsense understanding tasks showed smaller but consistently positive improvements, indicating that SFT provided stable enhancement to basic language understanding capabilities. These tasks test physics commonsense, context completion, and reference resolution abilities.

TruthfulQA-MC2 (+0.04%): The negligible improvement in factual accuracy indicates that SFT had limited effect on the model's fact-checking capabilities. This aligns with existing research suggesting that instruction fine-tuning has limited effectiveness in reducing model hallucinations, requiring specialized alignment training instead.

6 Conclusions & Analysis

6.1 Key Findings

This SFT experiment validated the following key findings:

1. The four-stage data cleaning effectively removed approximately 30.89% of low-quality samples, providing a high-quality data foundation for training;
2. Standard hyperparameter settings (learning rate 2e-5, batch size 64, 2 epochs) performed stably on the 0.5B model;
3. After SFT, the model achieved an average improvement of 5.59% across 8 benchmarks, with all tasks showing positive improvements, and knowledge reasoning tasks (MMLU, ARC) demonstrating the most significant gains;
4. The training process was stable with a declining loss curve decrease and no signs of overfitting;
5. TruthfulQA-MC2 showed negligible improvement (+0.04%), indicating that SFT has limited effectiveness in improving factual accuracy.

6.2 Limitations

The experiment had the following limitations:

1. The 0.5B model capacity was limited, constraining improvement space on complex reasoning tasks (ARC-Challenge);
2. Only a single GPU was used for training, without exploring multi-GPU parallel acceleration schemes;
3. Evaluation relied solely on automatic metrics, lacking manual evaluation to verify generation quality;
4. The minimal improvement on TruthfulQA-MC2 indicates that additional truthfulness alignment training is needed.

6.3 Future Directions

Future work could be improved in the following directions:

1. Increase training data scale to 500k-1M to explore the relationship between data volume and performance;
2. Try parameter-efficient fine-tuning methods such as LoRA to train larger models under the same memory constraints;
3. Add manual evaluation components to comprehensively assess generation quality and instruction-following capabilities;
4. Conduct alignment training such as DPO or RLHF to improve model factual accuracy and safety;
5. Conduct ablation studies to analyze the contribution of each data cleaning stage to final performance.